

Reproduction of karpathy/llama2.c: 15M-parameter Llama 2 trained on TinyStories

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kv:2610.00028v1 · Published 2026-10-08

Independent reproduction · Benchmark or leaderboard entry · Natural language processing

Abstract

Trained the README's smallest Llama 2 configuration (dim 288, 6 layers, 6 heads, about 15M parameters, Llama 2 tokenizer) on the pretokenized TinyStories dataset at the pinned commit with train.py defaults (batch 128, 4 gradient accumulation steps, bf16, torch.compile, seed 1337) for 15,000 iterations instead of the README's 20,000 to fit the time budget; the learning-rate schedule follows the shorter run. The exported model was then sampled with the C inference program run.c (256 tokens, temperature 0.8, seed 42). One NVIDIA RTX 5060 Ti.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Llama 2 15M on TinyStories	2026-10-08	1	1	Signed by the K-Veritas server

2 Results

r1: Llama 2 15M on TinyStories

Run	Metric	Final value	Step
1	train_loss	1.1897	14000
1	val_loss	1.1907	14000
1	c_inference_tokens_per_second	69.4935	

Run	Inline claim	Value	Phase
1	final_val_loss	1.1907	
1	c_inference_tokens_per_second	69.4935	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	8611.6 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	3dae219d9b292dd6af4174257c8ae16003eac2321b6f7344f02167d72b196109	6.9 MB
r1.kvbundle.zip	ea76c39ccd4d04686ac8a8d77aa09a8083de9812b35d1d3af6ca7b97d6db8cd3	61 KB

5 Links and licence

Reproduction of: <https://github.com/karpathy/llama2.c/tree/350e04fe35433e6d2941dce5a1f53308f87058eb>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: llama2, tinystories, language-model, inference.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (Llama 2 15M on TinyStories), kv:2610.00028v1/r1. Data hash d74a38cd96e3053227f568c286dcfa5f7a8d639d64e95e1b7cf8b32a2b5f00ab.

<https://kveritas.org/records/2610.00028v1/r1.pdf>